

DSAI 601 - Linear Regression

Fall 2026

Lecture: Mon/Thurs 10:00-12:00 or 1:00-3:00; SFD 529

Instructor: Robert Clements

Email: rclements [at] usfca [dot] edu

Office Hours: Mon 9:00-9:50; Thurs 3:00-3:50

Optional Textbook: Applied Linear Statistical Models by Kutner, Nachtsheim, Neter, and Li

Material Covered

We will cover the following topics, time permitting:

- Simple Linear Regression - Inference in Regression and Correlation Analysis - Multiple Regression and Matrix Notation - ANOVA and dummy variables - Model Selection, Model Validation, Diagnostics, and Remedial Measures - Logistic Regression

Course Learning Outcomes

By the end of this course, students will be able to:

- Understand the structural forms of classical simple and multiple linear and logistic regression models, as well as the assumptions underlying these models;
- Be familiar with the principles underlying, and results related to, parameter estimation for these classes of models;
- Formulate and test hypotheses using these basic models, as well as use them for both prediction and interpretation;
- Be able to use Python to load and clean data, fit regression models, and generate various outputs like ANOVA tables, confidence intervals for parameters, diagnostic assessments, etc.;
- Be able to check or to test whether or not diagnostics conform to the assumptions that underlie classical regression;
- Rigorously identify and manage outliers and influential observations and understanding the potential implications of leaving them uncensored;
- Be able to modify regression models to encourage fitted residuals to more closely conform to the assumptions that underlying classical regression, i.e., handle multicollinearity, heteroscedasticity, autocorrelation, a non-normal error term, and specification errors;
- Understand how to use dichotomous dependent variables, indicator (or dummy) independent variables, and interaction terms;
- Communicate the results of complete and well-reasoned regression analysis as an data analyst; and
- Critically evaluate and apply theoretical and applied concepts in statistical inference, honing your ability to engage with advanced statistical research and methodology.

Attendance and Professionalism

- Attendance is expected and required. If you are sick, you should **not** attend class. Let me know as soon as possible, before class, so that I can excuse you. If you wait until during or after class, you will not be excused. If you are sick multiple days and need to miss class, you must get a doctor's note for those absences to be excused. Assignments still must be turned in on-time.

Professionalism

- Be on time.
- Be respectful to me and others.
- Do not talk during class, it is distracting for everybody.
- Turn off notifications.
- I will not ask you to close your laptops in case you want to type notes, but, if I get complaints from students that your use of technology is unrelated to the course and distracting them during class, I will start to enforce a laptop ban.
- Cheating on quizzes, in any way, will result in automatically failing the course.
- Being unprofessional may result in a written warning. According to official program policy, two written warnings may result in expulsion from the program.

Homework

- You will be assigned computational and theoretical homework assignments to be completed and turned in on Canvas every Friday before midnight at 11:59pm Pacific time.
- Any homework turned in after the due date and time, but within 24 hours, will have 10% deducted from the grade. Homework turned in between 24 and 48 hours after the due date will have 20% deducted. Assignments turned in after 48 hours after the due date will not be graded and will earn a 0.
- Students are encouraged to work together on the assignments, but each student must turn in their own original work. If there is evidence that the work turned in is not original work, which includes copying another student's homework or using any solutions found online, all credit for that homework set will be forfeited.

Exams

- There will be 2 written exams, tentatively scheduled for: Thurs, Sept 17 and Tues, Oct 15. Details will be discussed in classes leading up to exam dates.
- No make-up or early exams will be given in order to ensure fairness and integrity of the class. Missing an exam without proper documentation of a personal illness or family emergency will result in a score of zero for that exam and will count towards your final grade. Any documentation must be submitted to the instructor before the exam in question at the student's earliest convenience. If the midterm is missed for a documented excused reason, the weight of the midterm will move to the final exam (i.e. the final exam will be worth the weight of the midterm + weight of the final). If the final exam score is higher than the midterm, the midterm grade is replaced by the final exam grade.
- Final exams cannot be made up unless due to an emergency. Notify me and the program Director as soon as possible before exam start time, and you will need to provide documenta-

tion (such as a doctor's note) and a makeup will be scheduled. Note that the makeup exam will be different in questions and format.

Closed book:

No cheat sheets allowed. A formula sheet will be provided.

Use of technology:

No technology is allowed.

On your desk:

Allowed on your desk: the quiz, pencil/pen/eraser, non-alcoholic drink.

Seating:

Seating will be assigned.

Leaving the room:

Bathroom breaks are allowed for one person at a time. You must raise your hand and get approval to go to the bathroom, and leave your devices (phone, laptop, smart watch) on the table. If I see that you are missing and I did not excuse you, I will take your quiz from your desk and grade it as-is.

Questions:

I do not answer any questions during the exam. If you are confused by a question, you must simply do your best. If you run out of space for your answer and need to write on the back or something of that sort, make sure to clearly label your paper.

Course Website

The class will be using Github to distribute all resources and Gradescope (<https://gradescope.com/>) to collect homework assignments. Exams will be uploaded to Gradescope. You will receive an email from the site allowing you to register. Please do not try to register on your own.

Grades

The grade breakdown is as follows:

Attendance/professionalism (5%)

Homework (20%)

Midterm Exam (30%)

Final Exam (45%)

Course Grade Cutoffs***:

A+ 97 - 100

A 93 - 97

A- 90 - 93

B+ 87 - 90

B 83 - 87

B- 80 - 83
C+ 77 - 80
C 73 - 77
C- 70 - 73
F 0 - 70

** Grading scheme is approximate. Instructor reserves the right to adjust grading scheme.

*** Cutoffs are approximate. Instructor reserves the right to alter any grade cutoffs. Final grade decisions will not be made until all assignments have been turned in and graded.

Regrades

If you have a legitimate appeal about a grade on an exam, please send me an email, with justification, within 5 days of receiving the graded exam. Do not approach me to verbally discuss it, I will not. Do not simply submit a regrade request in Gradescope, I may not even see it. If your appeal has merit, your entire exam will be completely re-graded, which means that your final grade on the exam may end up being lower, the same, or higher than the initial grade given. For simple mistakes your exam will not be re-graded, I'll just fix the issue.

University Policies

Credit-hour Policy

One unit of credit in lecture, seminar, and discussion work approximates one hour of direct faculty instruction (or 50 minutes plus a break) and a minimum of two hours of out-of-class student work per week through one 15-week semester. For further details, see USF's [Credit Hour Policy](#).

The below resources and additional information can be found in the [Student Life Resource Toolkit](#).

Students with Disabilities

The University of San Francisco is committed to providing equal access to students with disabilities. If you are a student with a disability, or if you think you may have a disability, please contact Student Disability Services (SDS) at sds@usfca.edu or (415) 422-2613, to speak with a disability specialist (please note all communication with SDS is private and confidential). If you are eligible for accommodations, please request that your accommodation letter be sent to me as soon as possible (students are encouraged to contact SDS at the beginning of the semester), as accommodations are not retroactive. Once I have been notified by SDS of your accommodations we can discuss your accommodations and ensure your access to this class or clinical setting. For more information please visit the [SDS website](#).

Behavioral Expectations

The University of San Francisco is committed to providing an environment consistent with the academic nature and core values of the institution in which students can participate in learning as a humanizing, social activity rather than a competitive exercise to support the full, integral development of each person and all persons, with the belief that no individual or group may rightfully prosper at the expense of others.

It is important you know what is expected of you academically and behaviorally through the published course syllabus, the [Student Conduct Code](#), and other instructions provided by the instructor. Therefore, you are expected to uphold the following:

- Behave in accordance with the Student Conduct Code and other USF policies
- Refrain from disrupting the ability of fellow students to learn or the instructor's ability to teach. Examples of disruption include:
 - Cell phone or computer use that significantly or repeatedly distracts others.
 - Coming to class late, leaving early, or excessively physically relocating oneself in the classroom.
 - Speaking frequently without being called on.
 - Yelling, cursing, or engaging in aggressive behavior.
- When interacting online, communicate in a respectful fashion. This includes, but is not limited to:
 - Refraining from name calling, profanity, or typing in all capital letters
 - Sending multiple emails with one sentence.
 - Avoiding rants or discussing non-relevant topics

Open discussion and disagreement are encouraged when done respectfully and in the spirit of academic discourse. There are also a variety of behaviors that, while not against a specific University policy, may create disruption in this course. Students whose behavior is disruptive or who fail to comply with the instructor may be dismissed from the class for the remainder of the class period and may need to meet with the instructor or Dean prior to returning to the next class period. If necessary, referrals may also be made to the Student Conduct process for violations of the Student Conduct Code.

Academic Integrity

As a Jesuit institution committed to *cura personalis*—the care and education of the whole person—USF has an obligation to embody and foster the values of honesty and integrity. All members of the USF academic community are responsible for maintaining the standards of honesty and integrity. The [honor code](#) applies to all students (undergraduate and graduate) in the College of Arts and Sciences, the School of Education, the School of Management, and the School of Nursing and Health Professions. Faculty and students in the School of Law should review their own [honor code](#) for policies and procedures. Students enrolled in distance learning (online courses) are subject to these policies as well as supplemental policies set forth by their program. All students should review and familiarize themselves with the honor code, prohibited conduct, and procedures.

Counseling and Psychological Services

Many college students experience mental health difficulties. Counseling and Psychological Services (CAPS) is a great source of support for addressing a range of mental health needs, including anxiety, stress, loneliness, relationship struggles, grief/loss, adjusting to change, identity development, racial/cultural concerns, and mild to moderate depression.

CAPS offers in-person and remote individual and group therapy to enrolled students. Students seeking services are scheduled for a 15-20-minute phone screening to assess immediate risk, as well as identify initial treatment needs, and make recommendations for next steps. These may

include a crisis or risk assessment, brief individual therapy (every 2-3 weeks), single session therapy, weekly group therapy, or referrals to off-campus providers. Please note, CAPS does not have a psychiatrist on staff to prescribe medication. Students in need of medication or with more complex mental health concerns are referred off-campus for specialized, comprehensive, treatment and/or ongoing, longer-term therapy. To make an appointment, call CAPS at 415.422.6352 or request an appointment via the [CAPS website](#). CAPS does not accept walk-in appointments.

Students with urgent mental health needs can call the CAPS All-Hours line (855-531-0761) to immediately speak with a mental health clinician. The CAPS All Hours line is available 24/7, including weekends/holidays, and accepts international calls, and should only be accessed for mental health crises. For all other mental health concerns, including scheduling, canceling, rescheduling, or leaving a message for a CAPS clinician, please call CAPS at 415.422.6352 during normal business hours.

All students are encouraged to check out the [CAPS website](#) and access our extensive online resources, podcasts, mental health apps, videos, self-care strategies, and more.

If you are concerned about a student and would like someone to follow up, please contact the Dean of Students Office at 415.422.5330. If you know someone who is an immediate risk of harming themselves or others, please contact Public Safety at 415.422.2911 in San Francisco, out of state or additional locations within California dial 911, or call the National Suicide & Crisis Lifeline by dialing 988.

Title IX

The Title IX Office seeks to stop, remedy, and prevent occurrences of sex and gender-based discrimination, sexual harassment, and sexual violence. The University has a [Policy](#) on Nondiscrimination Based on Sex and Gender, Sexual Harassment, and Sexual Misconduct for all Students, Employees, and Third-Parties. If you have experienced any behaviors related to this policy, we encourage you to report the incident. If you report these behaviors to any staff or faculty member, they must notify the USF Title IX Coordinator.

Students who wish to report any sexual misconduct should use the online mandatory reporting [form](#). You can also contact the Title IX Office directly by calling (415)422-4563, emailing titleix@usfca.edu, or visiting our office in Lone Mountain, Room 145.

As an employee at USF, and your Professor, I am a mandatory reporter, meaning I have to share any instances of sexual harassment or sexual violence shared with me or that become known to me. I will have to share this information, including names and any details known, to the Title IX Office to connect you with resources. If you would like more information about the resources available, you can ask me at any time this semester. You do not need to tell me why you are asking to get help for a friend, another student, or yourself.

If you are interested in accessing confidential resources, please consider these options. If you report any information to the following, it will not generate a report to the Title IX office:

- Students may speak to someone confidentially by contacting Counseling and Psychological Services at (415) 422-6352, or speaking to a clergy member in University Ministry at (415) 422-4463.

- To find more about reporting sexual misconduct (including sexual assault, relationship violence, or stalking) and resources at USF, visit the USF Title IX website.
- For off-campus resources, contact San Francisco Women Against Rape (SFWAR) at (415) 647-7273; www.sfwar.org

If during the course of the semester you need accommodations related to pregnancy or parenting, please let me know and I can connect you to the Title IX office. They are able to offer support through arranging accommodations, leave, or other support. The Title IX office is responsible for ensuring pregnant and postpartum students get the changes they need to stay healthy while being able to continue their education.

Confidential Resources for Reporting Sexual Misconduct

- Students may speak to someone confidentially which will not generate a report to the Title IX Office by contacting Counseling and Psychological Services at (415) 422-6352 during M-F 9-4pm, or speaking to a clergy member in University Ministry at (415) 422-4463.
- If you need to speak to a mental health clinician immediately, please call the CAPS 24/7 All Hours Line at 855-531-0761 (available daily, including weekends and holidays, and accepts international calls), Public Safety (415-422-2911), 911, the Suicide Hotline (dial 988), or go to your nearest emergency room.
- For off-campus resources and local Bay Area organizations check [here](#).

Learning, Writing, and Speaking Centers

The Learning, Writing, and Speaking Centers (LWSC) at USF provide individualized support to assist students in better understanding course material and to aid them on their path to success. Services are free and include tutoring, collaborative peer support services, academic skills coaching, writing, and speaking support. Services are available in-person and on Zoom.

LWSC staff can be reached Monday through Thursday between 9:00am-8:00pm and Friday between 9:00am-5:00pm at LWSC@usfca.edu or through the chatbox on [our myUSF webpages](#) or by phone at (415) 422-6713. To make an appointment for subject tutoring, academic skills coaching, the writing center, or the speaking center, students should visit the [Student Appointment Dashboard](#).

Communication

All course communications, like all other USF communications, will be sent to your USF official email address. You are therefore strongly encouraged to monitor that email account.

Gleeson Library

Looking for help with a research paper or project? Set up a consultation with a Librarian or get 24/7 research help online.

Use of Generative AI in This Course

The comprehensive use of generative AI tools is permitted with appropriate attribution.

In alignment with the [USF Honor Code](#), students must present work that is authentically their own. In this course at the instructor's discretion, the comprehensive use of generative AI tools is permitted with appropriate attribution. As this policy may be different from other course AI policies, please ask your instructor to clarify any questions you may have. To mitigate risks associated with the use of generative AI, please review the [Guidelines for Responsible Use of Artificial Intelligence \(AI\) at USF](#).

Note: I would, of course, discourage using generative AI to do your homework assignments. You will learn more by wrestling with the assignment yourself, and you may perform more poorly on the quizzes.